

Mathematics - K-12 Level Assessment 2

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

The first number in a number pattern is 28. The pattern rule is to add 14 to get the next number in the pattern. If the pattern continues, which statement is true?

- A. All the numbers in the pattern can be divided equally by 10.
- B. All the numbers in the pattern can be divided equally by 4.
- C. All the numbers in the pattern can be divided equally by 8.
- D. All the numbers in the pattern can be divided equally by 7.

Question 2

(2 marks)

Solve the equation $14 = w + 23$.

- A. -37
- B. -9
- C. 9
- D. 37

Question 3

(2 marks)

Erin made 12 pints of juice. She drinks 3 cups of juice each day. How many days will Erin take to drink all of the juice she made?

- A. 2 days
- B. 4 days

C. 8 days

D. 9 days

Question 4

(2 marks)

Mr. Jones rolls a six-sided cube numbered 1, 2, 3, 4, 5, 6. What is the probability he rolls a three?

A. 6-Jan

B. 5-Jan

C. 3-Jan

D. 2-Jan

Question 5

(2 marks)

If $f(x) = 2$ for all real numbers x , what is the value of $f(x + 2)$?

A. 2

B. 4

C. 0

D. 1

True / False

Question 1

(2 marks)

The sum of 365 and 56 is 311.

True

False

Question 2

(2 marks)

The number eighteen thousand ninety-six can be written in expanded form as $10\,000 + 8\,000 + 90 + 6$

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True

False

Question 3

(2 marks)

The inequality $x^2 - 6x + c < 0$ has real solutions for x when c is exactly 6.

True

False

Question 4

(2 marks)

The expression $-105 + (-14) + 34$ simplifies to -57.

True

False

Question 5

(2 marks)

The composition $f(g(h(x)))$ results in the expression $-15x^2 + 24$.

True

False

Long Form

Question 1

(10 marks)

Determine the equation of the tangent line to the curve defined by $F(t) = (t^2 + 1, 2^t)$ at the point where $y = 4$. Explain your reasoning and calculations.

Type your answer here...

Question 2

(10 marks)

Explain what a P-value represents in hypothesis testing, and analyze its significance in determining the validity of the null hypothesis in a statistical experiment.

Type your answer here...

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